

MH1300 FOUNDATIONS OF MATHEMATICS

Tutorial 7 Solutions

Ex. 4.5.21. Prove the statement if it is true and find a counterexample if it is false.

For all odd integers n , $\lceil n/2 \rceil = (n+1)/2$.

SOLUTION . Let n be any odd integer. [We must show that $\lceil n/2 \rceil = (n+1)/2$.]

Since n is odd, by definition, $n = 2k+1$ for some integer k . Substituting into the left-hand side of the equation to be proved gives

$$\left\lceil \frac{n}{2} \right\rceil = \left\lceil \frac{2k+1}{2} \right\rceil = \left\lceil k + \frac{1}{2} \right\rceil = k+1,$$

where $\lceil k + \frac{1}{2} \rceil = k+1$ by definition of ceiling because $k < k+1/2 < k+1$ and k is an integer.

On the other hand, substituting into the right-hand side of the equation to be shown gives

$$\frac{n+1}{2} = \frac{(2k+1)+1}{2} = \frac{2k+2}{2} = \frac{2(k+1)}{2} = k+1$$

also. Thus both the left- and right-hand sides of the equation to be proved equal $k+1$, and so both are equal to each other. In other words,

$$\lceil n/2 \rceil = (n+1)/2 \quad [\text{as was to be shown}].$$

□

Ex. 4.5.27. Prove the following statement.

For all real numbers x , if $x - \lfloor x \rfloor \geq 1/2$ then $\lfloor 2x \rfloor = 2\lfloor x \rfloor + 1$.

SOLUTION . Suppose x is any real number such that $x - \lfloor x \rfloor \geq 1/2$. Multiply both sides by 2 to obtain

$$2x - 2\lfloor x \rfloor \geq 1, \quad \text{or equivalently,} \quad 2x \geq 2\lfloor x \rfloor + 1.$$

Now by definition of floor, $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$. Hence

$$2x < 2\lfloor x \rfloor + 2.$$

Put the two inequalities involving x together to obtain

$$2\lfloor x \rfloor + 1 \leq 2x < 2\lfloor x \rfloor + 2.$$

By definition of floor, then,

$$\lfloor 2x \rfloor = 2\lfloor x \rfloor + 1.$$

□

Ex. 4.6.14. Prove the statement by contradiction.

For all prime numbers a, b , and c , $a^2 + b^2 \neq c^2$.

SOLUTION . Suppose not. That is, suppose there exist prime numbers a, b , and c such that $a^2 + b^2 = c^2$.

Subtracting b^2 from both sides of the equation gives that

$$a^2 = c^2 - b^2 = (c - b)(c + b).$$

Either $c - b = 1$ or $c - b > 1$ [since $c + b > 0$ and $a^2 > 0$, we must have $c - b > 0$].

Case 1: ($c - b = 1$.)

In this case, because both c and b are prime numbers and the only even prime number is 2, the only possible values for b and c are $c = 3$ and $b = 2$. Then

$$(c - b)(c + b) = 1 \cdot 5 = 5 = a^2,$$

and so $a = \sqrt{5}$. But this contradicts the assumption that a is a prime number.

Case 2: ($c - b > 1$.)

In this case, $a^2 = (c - b)(c + b)$ where both $(c - b) > 1$ and $(c + b) > 1$. Because a is prime, the only positive factors of a are 1 and a , and so, by the unique factorization theorem, the only positive factors of a^2 are 1, a , and a^2 .

Because both $(c - b)$ and $(c + b)$ are greater than 1, the only possibility is that both are equal to a . But this implies that $c - b = c + b$, which implies that $-b = b$, and hence that $b = 0$. This contradicts the supposition that b is a prime number.

Thus a contradiction is reached in both possible cases, and hence the supposition is false and the given statement is true. □

The **reciprocal** of a nonzero real number x is $1/x$.

Ex. 4.6.24. Prove the statement in two ways: (a) by contraposition and (b) by contradiction.

The reciprocal of any irrational number is irrational.

SOLUTION . (a) **Proof by contraposition:**

[We want to show $\forall x \in \mathbb{R} \setminus \{0\}$, if $1/x \in \mathbb{Q}$ then $x \in \mathbb{Q}$.]

Suppose x is a real number and $1/x$ is rational. [We must show that x is itself rational.]

Because $1/x$ is rational, there are integers a and b with $b \neq 0$ such that

$$1/x = a/b. \tag{*}$$

Note that $1/x$ is not zero, or else, x is not a number. Therefore, $a \neq 0$. Thus we may solve equation (*) for x to obtain $x = b/a$ where b and a are integers and $a \neq 0$. Hence, by definition of rational, x is rational [as was to be shown]. □

(b) **Proof by contradiction:**

Suppose not. That is, suppose $\exists$ a nonzero irrational number x such that $1/x$ is rational.

[We must show that this supposition leads logically to a contradiction.]

By definition of rational,

$$1/x = a/b, \tag{**}$$

where a and b are integers with $b \neq 0$. Note that $1/x$ is not zero, or else, x is not a number. Therefore, $a \neq 0$. Thus we may solve equation (**) for x to obtain $x = b/a$ where b and a are integers and $a \neq 0$. Hence, by definition of rational, $1/x$ is rational.

This is a contradiction to the supposition that x is irrational. [Hence the supposition is false and the statement is true.] $\square$

Remark: The converse and inverse statements are false. $\forall x \in \mathbb{R}$, if $x \in \mathbb{Q}$ then $1/x \in \mathbb{Q}$ is false since we can have a counterexample $x = 0$. $\square$

Ex. 4.7.18. The quotient-remainder theorem says not only that there exist quotients and remainders but also that the quotient and remainder of a division are unique. Prove the uniqueness. That is, prove that if a and d are integers with $d > 0$ and if q_1, r_1, q_2 , and r_2 are integers such that

$$a = dq_1 + r_1 \quad \text{where } 0 \leq r_1 < d$$

and

$$a = dq_2 + r_2 \quad \text{where } 0 \leq r_2 < d,$$

then

$$q_1 = q_2 \quad \text{and} \quad r_1 = r_2.$$

SOLUTION . Suppose that a and d are integers with $d > 0$ and that q_1, q_2, r_1 , and r_2 are integers such that

$$a = dq_1 + r_1 \quad \text{and} \quad a = dq_2 + r_2, \quad \text{where } 0 \leq r_1 < d \text{ and } 0 \leq r_2 < d.$$

[We must show that $r_1 = r_2$ and $q_1 = q_2$.]

Equating the right sides of both equations, we obtain

$$dq_1 + r_1 = dq_2 + r_2,$$

and so

$$r_2 - r_1 = dq_1 - dq_2 = d(q_1 - q_2). \tag{1}$$

Both r_1 and r_2 lie between 0 and d . Thus the difference of $r_2 - r_1$ lies between $-d$ and d , that is,

$$-d < r_2 - r_1 < d.$$

Substituting equation (1) into the inequalities, we arrive at

$$-d < d(q_1 - q_2) < d.$$

Since $d > 0$, dividing throughout the inequalities by d gives

$$-1 < q_1 - q_2 < 1.$$

Since $q_1 - q_2$ is an integer (being the difference of integers) and since it is strictly greater than -1 and strictly less than 1 , we conclude that $q_1 - q_2 = 0$. In other words, we must have $q_1 = q_2$.

Substituting $q_1 - q_2 = 0$ into equation (1), we get

$$r_2 - r_1 = d(q_1 - q_2) = d \cdot 0 = 0.$$

Therefore $r_1 = r_2$ also.

Hence, $r_1 = r_2$ and $q_1 = q_2$, as was to be shown. $\square$

Ex. 4.7.23. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

SOLUTION . Suppose not. Then $\sqrt{2} + \sqrt{3}$ is rational. [We must derive a contradiction.] By definition of rational,

$$\sqrt{2} + \sqrt{3} = \frac{a}{b} \quad (1)$$

for some integers a and b with $b \neq 0$. Also $a \neq 0$ because $a = b(\sqrt{2} + \sqrt{3})$ and both b and $\sqrt{2} + \sqrt{3}$ are nonzero.

Subtracting $\sqrt{2}$ from both sides of equation (1) gives

$$\sqrt{3} = \frac{a}{b} - \sqrt{2},$$

and squaring both sides of this equation yields

$$3 = \frac{a^2}{b^2} - 2\sqrt{2}\frac{a}{b} + 2.$$

Rearranging the terms and then simplifying gives

$$\begin{aligned} 2\sqrt{2}\frac{a}{b} &= \frac{a^2}{b^2} + 2 - 3 \\ &= \frac{a^2 - b^2}{b^2}. \end{aligned}$$

Multiplying both sides by $\frac{b}{2a}$, we arrive at

$$\sqrt{2} = \frac{a^2 - b^2}{2ab}. \quad (2)$$

(This is valid since $a \neq 0$.) From equation (2), we see that $\sqrt{2}$ may be expressed as a quotient of two integers $a^2 - b^2$ and $2ab$ and the integer $2ab$ is nonzero by the zero product property. Therefore, equation (1) shows that $\sqrt{2}$ is rational by definition of rational.

However, by Theorem 3.7.1, we know that $\sqrt{2}$ is irrational, and so a contradiction has been reached. [Hence the supposition is false and the statement is true.] $\square$

Ex. 4.7.24. Prove that $\log_5(2)$ is irrational.

SOLUTION . Suppose not. Then $\log_5(2)$ is rational. [We will show that this supposition leads to a contradiction.] By definition of rational,

$$\log_5(2) = \frac{a}{b}$$

for some integers a and b with $b \neq 0$.

By definition of logarithm, $5^{\frac{a}{b}} = 2$. Since $2 > 1$ we must have $5^{\frac{a}{b}} > 1$. This implies that $\frac{a}{b}$ is positive, or else $5^{\frac{a}{b}} \leq 1$. Therefore, we may assume that a and b are both positive. Raising both sides to the b -th power gives

$$5^a = 2^b.$$

Let $N = 5^a = 2^b$. Since $b \neq 0$, we know that $N > 2^0 = 1$. Thus we may consider the prime factorization of N . Because $N = 5^a$, the prime factors of N are all 5's. On the other hand, because $N = 2^b$, the prime factors of N are all 2's.

This contradicts the unique factorization theorem which states that the prime factors of any integer greater than 1 are unique except for the order in which they are written. Hence the supposition is false, and so $\log_5(2)$ is irrational. $\square$

Ex. 4.7.28. An alternative proof of the infinitude of the prime numbers begin as follows:

Proof: Suppose there are only finitely many prime numbers. Then one is the largest. Call it p . Let $M = p! + 1$. We will show that there is a prime number q such that $q > p$.

Complete this proof. ($p! = p(p-1) \cdots 3 \cdot 2 \cdot 1$.)

SOLUTION . Suppose there are only finitely many prime numbers. Then one is the largest. Call it p . Let $M := p! + 1$. We will show that there is a prime number q such that $q > p$. To see this, note that because all of the integers from 2 through p divide $p!$ and because

$$p! = M - 1,$$

then none of the integers from 2 through p divides M (by the same reasoning as in the proof of Proposition 4.7.3). But, by Theorem 4.3.4, some prime number q divides M . Since q cannot be any of $2, 3, \dots, p$, we have $q > p$.

Thus q is a prime number and $q > p$. This contradicts our hypothesis that p is the largest prime. Thus the supposition is false, and we conclude that there are infinitely many prime numbers. □